# Marking Scheme

2026 · Paper 1 · 105 marks

**FOR REVISION**

For revision and self-assessment only. This is not an official examination paper.

Difficulty rating：**4.7** / 5.0

**General Rules**
• M / A / PROOF mark types
• **f.t.**：承上題錯誤但方法正確者，可獲後續步驟分。
• Deduction cap： A(1)=1 · A(2)=1 · B=0 (u, pp)
• u = 單位遺漏或錯誤；pp = 準確度不符要求。每部分最多扣 1 分。
• 答案正確且步驟合理者可獲滿分。
• 符號書寫略有出入但意思清楚者不扣分。 · Benefit of doubt
• 分數、小數或根式形式均接受，除非題目另有指明。

## Section A(1) (35 marks)

### 1. Laws of Indices

**a** (3 marks)

M1 $(x^2y^{-3})^3 = x^6y^{-9}$

M1 $\dfrac{x^6y^{-9}}{x^{-5}y^2} = x^{11}y^{-11}$

A1 $= \dfrac{x^{11}}{y^{11}}$

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### 2. Algebraic Fractions

**a** (3 marks)

M1 $= \dfrac{5(x-1)-2(2x+3)}{(2x+3)(x-1)}$

M1 $= \dfrac{5x-5-4x-6}{(2x+3)(x-1)}$

A1 $= \dfrac{x-11}{(2x+3)(x-1)}$

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### 3. Rates and Ratios

**a** (3 marks)

M1 $Let a = 4k and b = 5k$

M1 $12k + 10k = 110 \Rightarrow k = 5$

A1 $b = 25$

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### 4. Transformations in the Coordinate Plane

**a** (1 mark)

A1 $A' = (-2, -5)$

**b** (2 marks)

M1 $Slope of BC = 2; \dfrac{3-(-5+k)}{1-(-2)} = 2$ f.t.

A1 $k = 2$

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### 5. Factorization

**a** (1 mark)

A1 $4b(a-3)$

**b** (1 mark)

A1 $(a-3)(a+3)$

**c** (2 marks)

M1 $=(a-3)(a+3)-4b(a-3)$ f.t.

A1 $=(a-3)(a+3-4b)$

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### 6. Linear Inequalities in One Unknown

**a** (3 marks)

M1 $x+5<2 \Rightarrow x<-3$

M1 $-5x \geq 25 \Rightarrow x \leq -5$

A1 $x<-3$

**b** (1 mark)

A1 $-4$ f.t.

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### 7. Percentages

**a** (2 marks)

M1 $Markedprice \times (1-15\%) = 1020$

A1 $Markedprice = \1\,200$

**b** (2 marks)

M1 $Cost \times (1+60\%) = 1200$ f.t.

A1 $Cost = \750$ f.t.

**c** (1 mark)

A1 $Newsellingprice = 1200(1-40\%) = \720 < \750, sothereisaloss; no.$ f.t.

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### 8. Congruent and Similar Triangles

**a** (2 marks)

M1 $AE = DE(given), \angle BAE = \angle CDE(alt.\ \angles, AB \parallel CD), \angle AEB = \angle DEC(vert.\ opp.\ \angles)$

A1 $\therefore \triangle ABE \cong \triangle DCE(ASA)$

**b** (3 marks)

M1 $CD = AB = 12cm(corr.\ sides, \cong \triangles), soCG = 12-8 = 4cm$ f.t.

M1 $\triangle ABH \sim \triangle GCH; \frac{CH}{BH} = \frac{GH}{AH} = \frac{CG}{AB} = \frac{1}{3}, soCH = 5cm, GH = 4cm$ f.t.

A1 $Perimeter = 4+5+4 = 13cm$ f.t.

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### 9. Measures of Dispersion

**a** (2 marks)

M1 $8dataarelessthan2and13dataaregreaterthan2; need13 < \frac{21+k}{2}(orequivalent)$

A1 $Leastpossiblek = 6$

**b** (1 mark)

A1 $Modeis3, sok < 9; greatestpossiblek = 8$

**c** (2 marks)

M1 $Standarddeviationsfork = 6,7,8areabout1.227, 1.206, 1.186$ f.t.

A1 $Greatestpossibles.d. \approx 1.23$ f.t.

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**Section A(2) (35 marks)**

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## 10. Polynomials

**a** (3 marks)

M1 $f(-3) = -54 + 9a - 3b - 6 = 0 \Rightarrow 3a - b = 20$

M1 $f(2) = 16 + 4a + 2b - 6 = 60 \Rightarrow 2a + b = 25$

A1 $a = 9, b = 7$

**b** (3 marks)

M1 $f(x) = (x+3)(2x^2 + 3x - 2)$ f.t.

M1 $= (x+3)(2x-1)(x+2)$ f.t.

A1 $The roots are -3, \frac{1}{2} and -2, all rational; I agree.$ f.t.

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## 11. Variations

**a** (3 marks)

M1 $p(x) = ax + bx^2, where a, b are non-zero constants$

M1 $a + b = -3 and 3a + 9b = 3$

A1 $p(x) = 2x^2 - 5x$

**b** (3 marks)

M1 $2x^2 - 5x - c = 0$ f.t.

M1 $\Delta = 25 + 8c > 0$ f.t.

A1 $c > -\frac{25}{8}$ f.t.

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## 12. Measures of Dispersion

**a** (3 marks)

M1 $Range = 89 - 52 = 37 kg; Q_1 = \frac{58 + 60}{2} = 59 kg$

M1 $Q_3 = \frac{(70 + k) + 84}{2}; 37 - \left(\frac{154 + k}{2} - 59\right) = 16$

A1 $k = 6$

**b(i)** (2 marks)

M1 $Before training, Q_3 = \frac{76 + 84}{2} = 80 kg$ f.t.

A1 $The upper quartile decreases by 80 - 74 = 6 kg$ f.t.

**b(ii)** (2 marks)

M1 $Before: range = 37 kg, inter-quartile range = 21 kg; after: range = 35 kg, inter-quartile range = 17 kg$ f.t.

A1 $Both measures are smaller after training, so yes.$ f.t.

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## 13. Locus and Coordinate Geometry

**a** (1 mark)

A1 $\Gamma is the perpendicular bisector of AB$

**b** (3 marks)

M1 $Let A = (a, 0), B = (0, b); AB \perp \Gamma: -\frac{b}{a} \times 2 = -1 \Rightarrow a = 2b$ f.t.

M1 $Mid-point \left(\frac{a}{2}, \frac{b}{2}\right) lies on \Gamma: a - \frac{b}{2} - 6 = 0$ f.t.

A1 $b = 4, a = 8; L: x + 2y - 8 = 0$

**c** (3 marks)

A1 $C = (3, 0), M = (4, 2)$ f.t.

M1 $Area = area of \triangle OAB - area of \triangle CAM = \frac{1}{2}(8)(4) - \frac{1}{2}(5)(2)$ f.t.

A1 $= 11$ f.t.

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### 14. Areas and Volumes of Solids

**a** (3 marks)

M1 $Small cone: base radius 6cm, height 8cm$

M1 $Volume = \frac{1}{3}\pi(12^2)(16) - \frac{1}{3}\pi(6^2)(8)$

A1 $= 672\pi cm^3$

**b** (3 marks)

M1 $Slant heights: \sqrt{12^2+16^2} = 20cm and 10cm$

M1 $Curved surface area = \pi(12)(20) - \pi(6)(10) = 180\pi cm^2$

A1 $Total surface area = 180\pi + 144\pi + 36\pi = 360\pi cm^2$ f.t.

**c** (3 marks)

M1 $Side of cube = \sqrt[3]{672\pi} \approx 12.83cm$ f.t.

M1 $Total surface area of cube = 6(12.83)^2 \approx 987cm^2; 360\pi \approx 1131cm^2$ f.t.

A1 $987 < 1131, so yes.$ f.t.

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## Section B (35 marks)

### 15. Probability

**a** (2 marks)

M1 $\frac{C_3^4 + C_3^3 + C_3^5}{C_3^{12}}$

A1 $= \frac{15}{220} = \frac{3}{44}$

**b** (2 marks)

M1 $\frac{4 \times 3 \times 5}{C_3^{12}}$

A1 $= \frac{3}{11}$

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### 16. Logarithms

**a** (2 marks)

M1 $\log_4(x^2) = \frac{2\log_2 x}{\log_2 4} = \log_2 x, so \log_2 x = 1 + u$

A1 $(1+u)u = 6 \Rightarrow u^2 + u - 6 = 0$

**b** (2 marks)

M1 $(u-2)(u+3) = 0, u = 2 or -3; \log_2 x = 3 or -2$ f.t.

A1 $x = 8 or x = \frac{1}{4}$ f.t.

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### 17. Arithmetic and Geometric Sequences

**a** (3 marks)

M1 $Let the first term be a; the terms are a, a+4, a+16$

M1 $(a+4)^2 = a(a+16)$

A1 $8a + 16 = 16a \Rightarrow a = 2$

**b** (3 marks)

M1 $\frac{n}{2}[2(2) + 2(n-1)] = n(n+1) > 1\,000\,000$ f.t.

M1 $999 \times 1000 = 999\,000 < 1\,000\,000 and 1000 \times 1001 > 1\,000\,000$ f.t.

A1 $Least n = 1000$ f.t.

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## 18. Quadratic Functions and Coordinate Geometry

**a** (2 marks)

M1 $f(x) = (x-k)^2 + 3k$

A1 $V = (k, 3k)$

**b(i)** (4 marks)

A1 $W = (-k, 3k)$ f.t.

M1 $VW is horizontal, so the altitude from P is the y-axis; let P = (0, p)$ f.t.

M1 $OV \perp WP : 3 \times \dfrac{p-3k}{0-(-k)} = -1$ f.t.

A1 $P = \left(0, \dfrac{8k}{3}\right)$ f.t.

**b(ii)** (3 marks)

M1 $The centre lies on the perpendicular bisector of VW (y-axis), which contains P and Q, so PQ is a diameter$ f.t.

M1 $\angle PVQ = 90° : (k)(k) + \left(3k - \dfrac{8k}{3}\right)(3k-12) = 0$ f.t.

A1 $2k^2 - 4k = 0, k = 2 (since k > 0)$ f.t.

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## 19. Equations of Circles and Tangents

**a** (4 marks)

M1 $x^2 + \left(\dfrac{35-3x}{4}\right)^2 - 4x - 2\left(\dfrac{35-3x}{4}\right) + c = 0$

A1 $25x^2 - 250x + (945 + 16c) = 0$

M1 $\Delta = 250^2 - 4(25)(945+16c) = 0$ f.t.

A1 $c = -20$ f.t.

**b(i)** (2 marks)

M1 $G = (2,1), radius = 5, PG = 9; \sin \angle APG = \dfrac{5}{9}$ f.t.

A1 $\cos \angle APB = 1 - 2\left(\dfrac{5}{9}\right)^2 = \dfrac{31}{81}$ f.t.

**b(ii)** (2 marks)

M1 $PA = \sqrt{9^2 - 5^2} = 2\sqrt{14}; \dfrac{AB}{2} = PA \sin \angle APG = 2\sqrt{14}\left(\dfrac{5}{9}\right)$ f.t.

A1 $AB = \dfrac{20\sqrt{14}}{9}$ f.t.

**b(iii)** (4 marks)

M1 $\angle PAG = \angle PBG = 90°, so PG is a diameter of the circumcircle; circumradius = \dfrac{9}{2}$ f.t.

M1 $Distance from P to AB = PA \cos \angle APG = \dfrac{56}{9}; area of \triangle PAB = \dfrac{1}{2}\left(\dfrac{20\sqrt{14}}{9}\right)\left(\dfrac{56}{9}\right) = \dfrac{560\sqrt{14}}{81}$ f.t.

A1 $In-radius = \dfrac{\text{area}}{\text{semi-perimeter}} = \dfrac{560\sqrt{14}/81}{28\sqrt{14}/9} = \dfrac{20}{9}$ f.t.

A1 $\dfrac{9}{2} = \dfrac{81}{18} > \dfrac{80}{18} = 2 \times \dfrac{20}{9}; I agree.$ f.t.

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